| VISION | MISSION |
|---|---|
| To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering. | 1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.<br>2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.<br>3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering. |

# Practical No.-0

**Aim:** Introduction to Design and analysis of Algorithm lab.

**Theory:**

## Algorithm

A finite set of instruction that specifies a sequence of operation is to be carried out in order to solve a specific problem or class of problems is called an Algorithm.

There are several strategies for any algorithm's design and evaluation, including brute force algorithms, divide and conquer algorithms, dynamic programming, and greedy algorithms. Each method has its very own strengths and weaknesses, and the choice of approach depends on the nature of the problem being solved.

Algorithm analysis is often performed by examining the algorithm's worst-case time and space complexity. The time complexity of an algorithm refers to the amount of time it takes to clear up a problem as a characteristic of the input size. The space complexity of an algorithm refers to the quantity of memory required to solve a problem as a function of the enter length.

**Need of Algorithm**

1. To understand the basic idea of the problem.
2. To find an approach to solve the problem.
3. To improve the efficiency of existing techniques.
4. To understand the basic principles of designing the algorithms
5. To compare the performance of the algorithm with respect to other techniques.
6. It is the best method of description without describing the implementation detail.
7. The Algorithm gives a clear description of requirements and goal of the problem to the designer.
8. A good design can produce a good solution.
9. To understand the flow of the problem.

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10. To measure the behavior (or performance) of the methods in all cases (best cases, worst cases, average cases)

11. With the help of an algorithm, we can also identify the resources (memory, input-output) cycles required by the algorithm.

12. With the help of algorithm, we convert art into a science.

13. To understand the principle of designing.

14. We can measure and analyze the complexity (time and space) of the problems concerning input size without implementing and running it; it will reduce the cost of design.

## Analysis of algorithm

The analysis is a process of estimating the efficiency of an algorithm. There are two fundamental parameters based on which we can analysis the algorithm:

- **Space Complexity:** The space complexity can be understood as the amount of space required by an algorithm to run to completion.
- **Time Complexity:** Time complexity is a function of input size **n** that refers to the amount of time needed by an algorithm to run to completion.

### Algorithm Design Techniques

The following is a list of several popular design approaches:

**1. Divide and Conquer Approach:** It is a top-down approach. The algorithms which follow the divide & conquer techniques involve three steps:

- Divide the original problem into a set of subproblems.
- Solve every subproblem individually, recursively.
- Combine the solution of the subproblems (top level) into a solution of the whole original problem.

**2. Greedy Technique:** Greedy method is used to solve the optimization problem. An optimization problem is one in which we are given a set of input values, which are required either to be maximized or minimized (known as objective), i.e. some constraints or conditions.

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- Greedy Algorithm always makes the choice (greedy criteria) looks best at the moment, to optimize a given objective.
- The greedy algorithm doesn't always guarantee the optimal solution however it generally produces a solution that is very close in value to the optimal.

**3. Dynamic Programming:** Dynamic Programming is a bottom-up approach we solve all possible small problems and then combine them to obtain solutions for bigger problems.

This is particularly helpful when the number of copying subproblems is exponentially large. Dynamic Programming is frequently related to **Optimization Problems**.

**4. Branch and Bound:** In Branch & Bound algorithm a given subproblem, which cannot be bounded, has to be divided into at least two new restricted subproblems. Branch and Bound algorithm are methods for global optimization in non-convex problems. Branch and Bound algorithms can be slow, however in the worst case they require effort that grows exponentially with problem size, but in some cases we are lucky, and the method coverage with much less effort.

**5. Randomized Algorithms:** A randomized algorithm is defined as an algorithm that is allowed to access a source of independent, unbiased random bits, and it is then allowed to use these random bits to influence its computation.

**6. Backtracking Algorithm:** Backtracking Algorithm tries each possibility until they find the right one. It is a depth-first search of the set of possible solution. During the search, if an alternative doesn't work, then backtrack to the choice point, the place which presented different alternatives, and tries the next alternative.

**7. Randomized Algorithm:** A randomized algorithm uses a random number at least once during the computation make a decision.

**Conclusion:** Thus, I have studied the introduction to Design and analysis of Algorithm.

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